# Encoders vs PsyDefDetect@BioNLP2026
# 3-cfu Project Work

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## Abstract

Psychological interpretation of an individual's words has always required a great capacity for adaptation and pattern recognition. This is precisely what the PsyDefDetect@BioNLP2026 competition is aiming for, in which the goal is to identify and classify psychological defense mechanisms in emotional support dialogues. Although the use of Large Language Models is the most immediate choice for psychological pattern recognition, the interest of this project shifts towards the use of Encoder-only architectures. Each of those considered is tested using different models, among which the best performing is mental/mental-bert-base-uncased , which corresponds to BERT-base trained with mental health-related posts collected from Reddit. Despite the gap with the top positions, the proposed architectures show real potential for use, potential that can be tested in less computationally and financially limited environments.

## 1 Introduction

Psychological defenses are the "immune system" of the mind, shaping what speakers disclose and how they accept or resist help. Despite their critical role in mental health and counseling, defensive functioning remains largely unmodeled in current emotional support conversation systems. The goal of PsyDefDetect@BioNLP2026 (Na et al., 2026) is to develop models that can recognize subtle, context-dependent defensive maneuvers. The most immediate solution to this type of problem involves using Large Language Models, doing zero/few-shot prompting. They have more reasoning capabilities but with penalties they lose sustainability and increase computational effort. However, my choice lies in considering only architectures based on Encoder-only models, to study whether it is possible to maintain high performance with lighter and more robust architectures. A wide variety of classification structures and methodologies were tested during the project to try to achieve optimal results. Although good future prospects are observed, the biggest obstacle is attributed to data separability, rather than architectural choices.

## 2 Background

- Psychological defense mechanisms

Psychological defense mechanisms, a core concept from psychoanalytic theory, are unconscious processes that individuals employ to reduce anxiety and maintain psychological equilibrium when facing internal conflicts or external stressors. These mechanisms profoundly influence an individual's emotions, cognitions, and behaviors, holding significant implications for mental health. On this basis, J. Christopher Perry builds a hierarchical structure that places psychological defenses along a maturity scale, starting from a mature and conscious level to a conflicted and impulsive one (Perry and Henry, 2004) (Di Giuseppe and Perry, 2021). The classification task is based on this particular scale.

- MentalBERT

A model can be trained and specialized on a specific vocabulary to better capture nuances and syntactic relationships that a general model may not be able to fully represent. MentalBERT is an example: it was obtained by training a BERT-based model on a corpus of text, extracted from Reddit posts related to mental health (Ji et al., 2021).

- Multiple Instance Learning

Multiple Instance Learning (MIL) (Ilse et al., 2018) is a supervised learning paradigm where a small group of data is grouped and labeled. In this project, one of its variants, the MIL Embedding Level, is presented. Based on this, instances of a single dialogue are grouped into a single set, and their embeddings are then aggregated using Gated Attention, resulting in a representation of the entire dialogue;then this representation can be classified.

- Gated Attention

Gated Attention (Ilse et al., 2018) is a mathematical operation that, thanks to the use of two nonlinear functions (usually a tangent and a sigmoid), can be used to assign an importance score to the data of a set (such as a set of embeddings).

- Multitask Learning

Multitask Learning (Liebel and Körner, 2018) is a paradigm in which model parameters are updated by minimizing a loss that depends on more than one task. It includes Auxiliary Task Learning, a variant in which there is a main task supported by auxiliary tasks that aim to optimize the training phase to improve the performances on the main task, and are finally removed during the inference phase.

## 3 System description

During the project, I set out to build two structurally different architectures simply because, since it was a very complex task, taking different paths would give me a better chance of achieving good results.

As mentioned above, the foundation is a pre-trained encoder that is used to create the embeddings that are given as input to the architectures. It should be noted that, with regard to architectural choices such as the encoder model, the classifier and with regard to hyperparameters, many combinations have been tested to arrive at the results obtained.

Architectures:

- Multitask Distance Learning

The architecture is based on the assumption that within the 9 classes there are subsets of classes that are similar to each other, subsets that belong to different macro-categories. There is a first input processing stage where the seeker turns are concatenated and passed to mentalBert. It computes the relevant embeddings and mean pooling is used to obtain the embedding vector that will be given as input to the classifiers. There are 5 different linear heads, one is the main one that considers all 9 classes, the other 4 are auxiliary binary heads that have the purpose of establishing whether or not the sample belongs to one of the following subsets: "is it a defense at all?", "is the defense about image distorting?", "is the defense method cognitive?", "is the defense method mature?" The loss of the auxiliary heads is added to the loss of the main head, which allows the encoder to better understand the different psychological aspects to consider when classifying a psychological defense case. The auxiliary heads are then removed during inference. To better understand the different semantic distance between classes, a distance matrix is introduced. It corresponds to a 9x9 matrix where the value of cell $(i, j)$ is a digit from 0 to 3 which indicates how semantically distant class $i$ is from class $j$. The final loss used in the training phase corresponds to the weighted sum of the 5 loss entropy and the distance loss.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda_d \times \mathcal{L}_{\text{dist}} + \lambda_a \times \sum_{k \in \{A,B,C,D\}} \mathcal{L}_{\text{BCE}}^{(k)}$$

Only the last two layers of the encoder are fine-tuned, thus maintaining the basic syntactic rules and adjusting only the most complex representations.

- Hierarchical MIL

The idea behind the structure is that it is difficult to classify 9 classes at once. The decision is decomposed into a hierarchical structure where there are 4 different classifiers (see figure 2), allowing each to specialize on a narrower and more balanced subproblem. The encoder remains frozen and is common to all nodes, which have the same basic structure. The first node is used to distinguish between defense, non-defense, and ambiguous cases; the second operates on defensive classes, being able to identify three individual classes + two subgroups. The first of these subgroups, relating to image-distorting cases, is classified by the third node, while the second subgroup, relating to reality avoidance cases, is classified by the fourth and final node. The nodes have a common structure, inside there is a Gated Attention Layer that takes care of taking the bag of turn embeddings (in this case the last 3 seeker's turns) and makes them weighted pooling, obtaining a single dialogue embedding. Each dialogue embedding thus obtained can pass through to a linear node classifier. This way, a single dialog can traverse up to 3 nodes to be classified, so individual turn embeddings are pre-computed to make the pipeline more efficient.

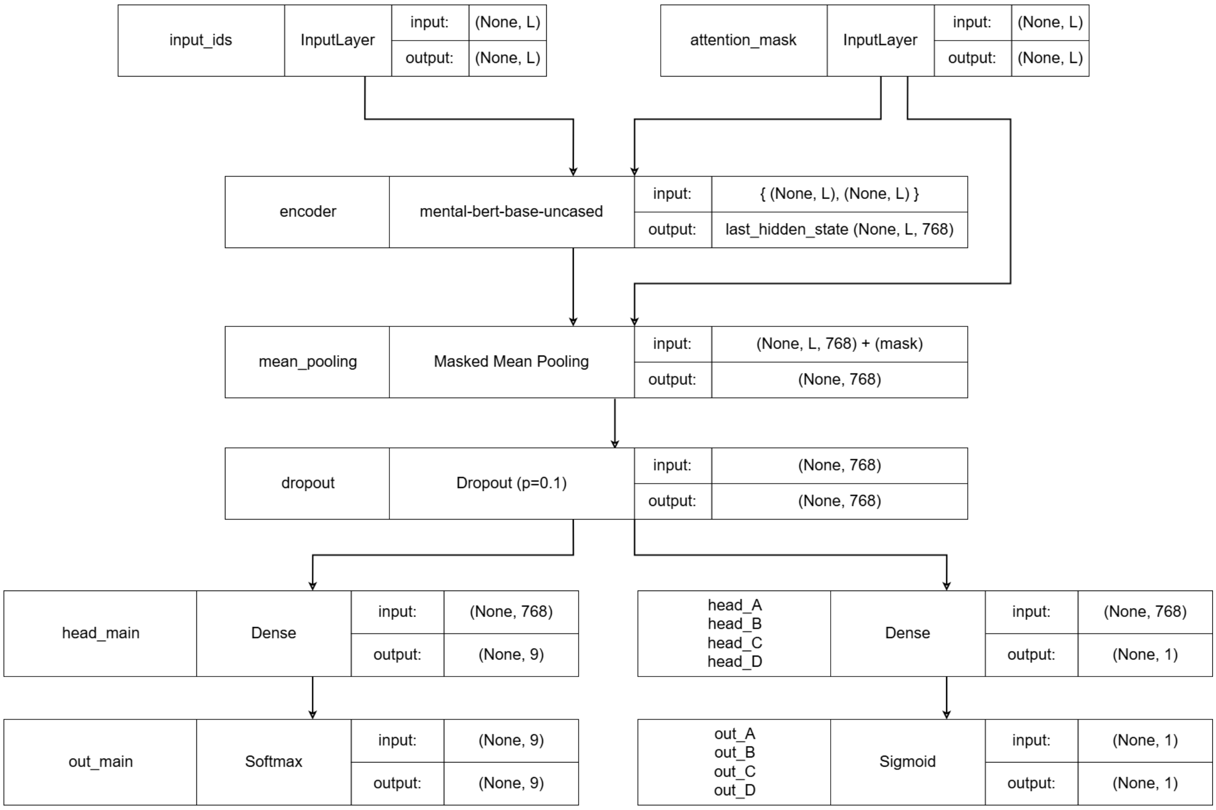


Figure 1: Multitask learning distance architecture

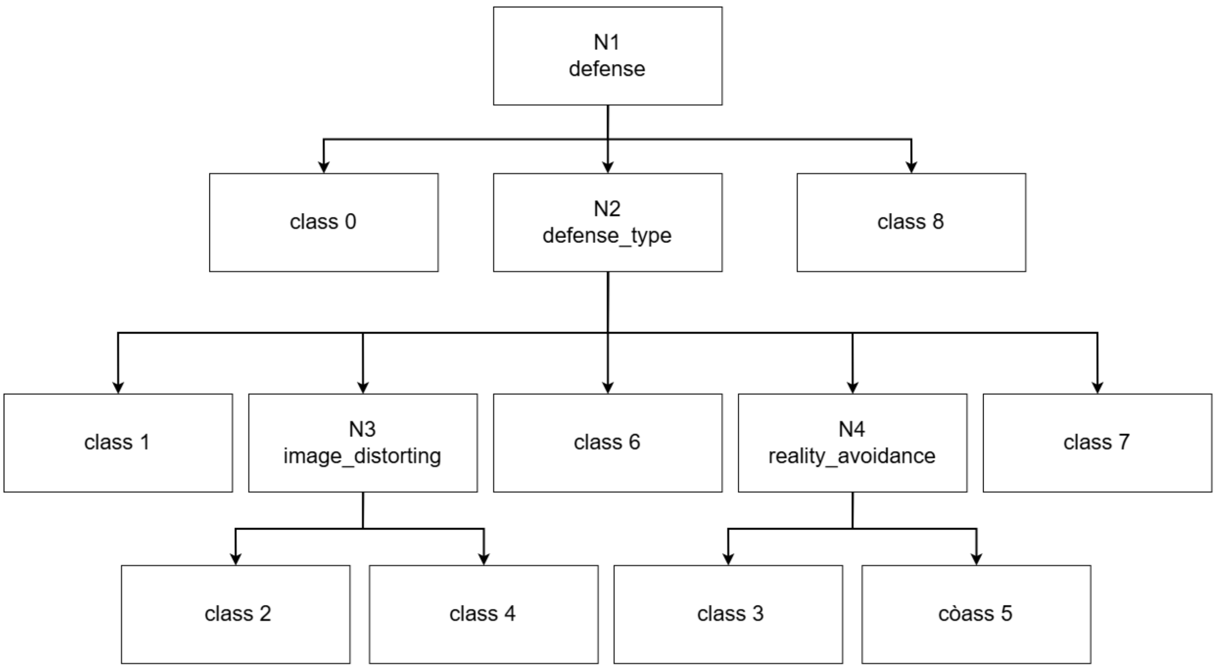


Figure 2: Hierarchy of the classification workflow.

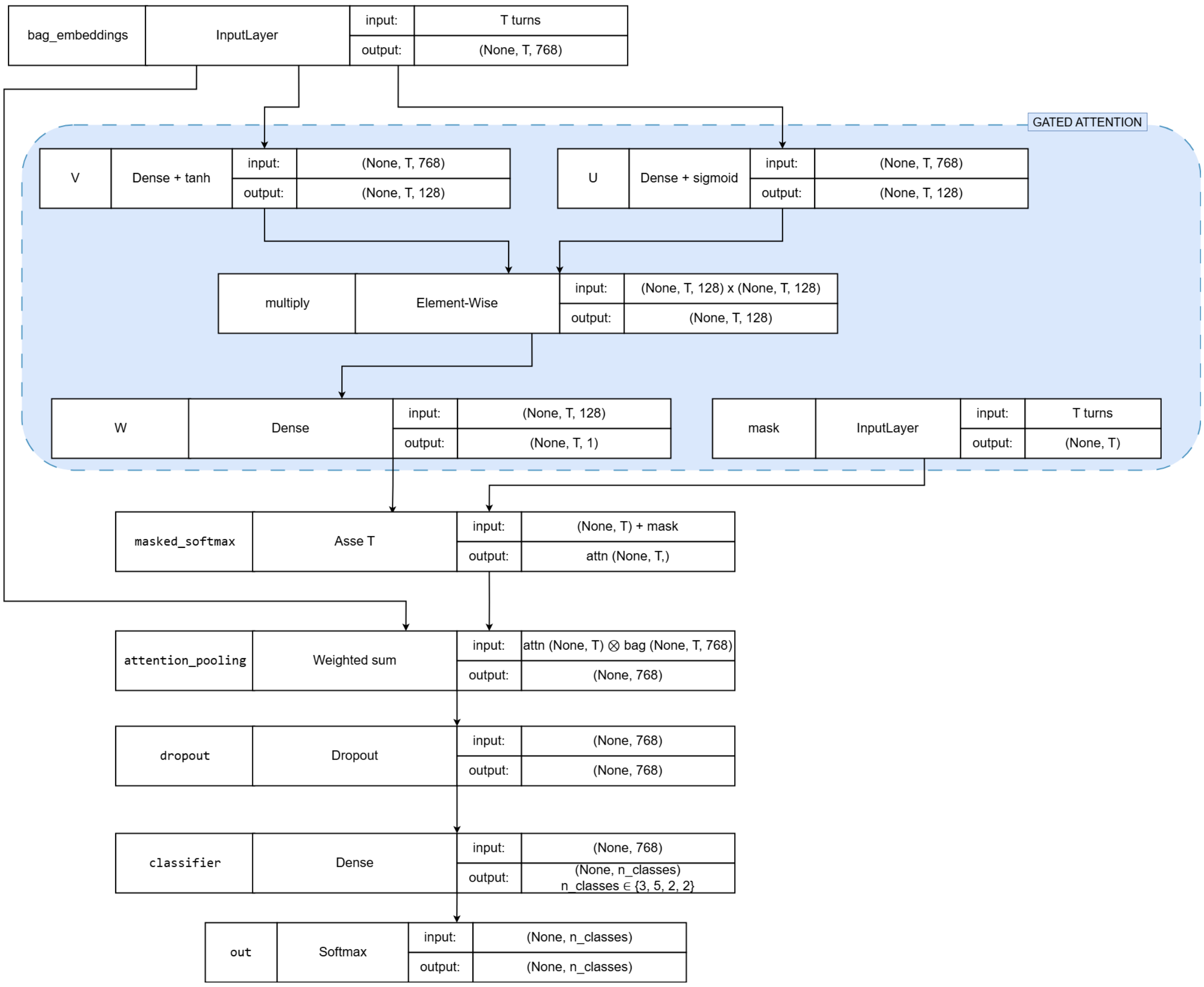


Figure 3: Hierarchical nodes architecture

## 4 Data

The task organizers directly provided the dataset to be used, which corresponds to PSYDEFCONV (Na et al., 2025), a conversational dataset annotated with defense levels based on the DMRS. The dataset is constructed from a stratified subset of the ESConv corpus (Liu et al., 2021) to ensure different coverage of problem types and emotions.

The corpus contains 200 dialogues and 4,709 total utterances, including 2,336 help-seeker turns annotated for defense levels. Each dialogue consists of a back-and-forth between a seeker and a supporter. Both the training set and the test set are highly unbalanced (see Figure 4), but remember that, since the task was presented in the form of a competition, the test set was released later.

Initially, a balanced validation set was extracted by choosing to maintain 15 samples per class. Prompt-based synthetic data generation is then done using 'Opus 4.6' as the Large Language Model generator, passing a custom prompt and the handbook provided within the task. The artificial dialogues were grouped within a CSV file that is merged with the existing dataset. Back-translation (Sennrich et al., 2016) was also tested using several Latin and non-Latin languages, but after initial tests it was seen that the results worsened, probably because with each translation, interpretation errors accumulated, modifying the interpretation of expressions and generics compared to the original intent. For this reason it was finally discarded as a data augmentation method. Furthermore, at the beginning of the project, it was also considered as an option to perform data reduction on class 7 to make the dataset much more balanced. The matrix dialogues were converted into a TF-IDF matrix, then through a clustering of 200 clusters it was possible to choose the sample closest to each centroid, obtaining 200 samples The idea was to capture as many language samples as possible to maintain most of the variance within the class. This methodology was also later shelved due to the worsening of the results that followed.

## 5 Experimental setup and results

The experiments saw the use of macro-F1 as a reference metric following the directives of the competition itself, the splits used are: training augmented with the LLM-generated samples, validation built in the pre-processing phase, test taken as it is by the competition task once published. The code uses the Huggingface framework to work with models. All experiments were performed using free Google Colab, specifically the models ran on an NVIDIA Tesla T4 with 16 GB of VRAM (GDDR6) memory. The hyperparameters were chosen through manual search, going by trial and error and evaluating the results as the results were evaluated to see what could be changed to improve the classifications. In both models were used: optimizer AdamW, weight_decay=$10^{-4}$, batch_size=16, early_stopping on macro-F1 of the validation set with a best-checkpoint model selection, and as encoder 'mental-bert-base-uncased

- Multitask Distance Architecture

Only the last two layers of the encoder were not frozen, lr=$2 \times 10^{-5}$, dropout=0.1, max_epochs=20, grad_clip=1.0. A first version of the psychological distance matrix was defined by passing the handbook to an LLM model, after which the distances were manually updated considering the confusion matrices obtained during validation. The loss final hyperparameters are $\lambda_d = 0.1$ and $\lambda_a = 0.5$).

| **Loss composition** | **Val** | **Test** | **$\Delta$** |
|---|---|---|---|
| Cross-Entropy | 0.2721 | 0.2254 | 0.0467 |
| CE + distance | 0.3068 | 0.2383 | 0.0685 |
| CE + auxiliary | 0.3061 | 0.2513 | 0.0548 |
| CE + dist + aux | **0.2899** | **0.2539** | **0.0360** |

Table 1: Architecture results as loss composition varies

As we can see in the results, distances and auxiliary losses not only help the model achieve higher scores, their combination also help make the model more stable and less prone to overfitting.

- Hierarchical Architecture

The encoder is completely frozen, each node has the following hyperparameters: hidden_dim=128, dropout=0.1, lr=$10^{-3}$, class-weighted cross entropy loss. In this architecture, having the possibility to pre-compute embeddings allows for a larger hyperparameter search.

As in the previous architecture, in this one too the mental-bert-base-uncased encoder turns out to be the one that best captures the psychological meaning of sentences.

In figure 5 it is possible to see an example of misclassification where the model recognizes a defense system even though it is not there.

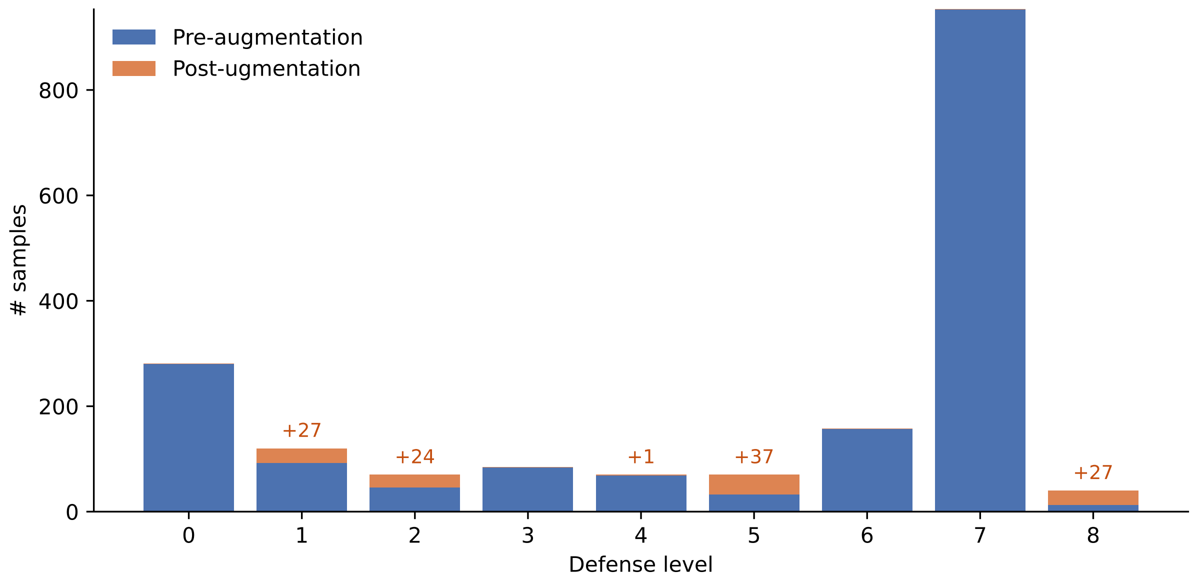


Figure 4: Training data distribution

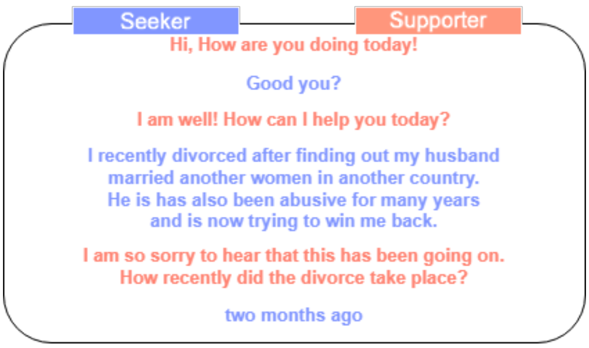


Figure 5: Example of a sample with label 0 being misclassified as 2

| class | Multi | Hier |
|---|---|---|
| 0 | 0.65 | 0.79 |
| 1 | 0.31 | 0.09 |
| 2 | 0.13 | 0.20 |
| 3 | 0.10 | 0.14 |
| 4 | 0.13 | 0.09 |
| 5 | 0.07 | 0.03 |
| 6 | 0.36 | 0.15 |
| 7 | 0.55 | 0.62 |
| 8 | 0.00 | 0.20 |
| Total | 0.25 | 0.26 |

Table 3: Architecture F1 per-class as the encoder used varies

| Encoder | Test |
|---|---|
| bert-base-uncased | 0.2343 |
| mental-bert-base-uncased | **0.2580** |
| mental-bert-base-cased | 0.2189 |
| mental-roberta-base | 0.2263 |
| Bio_ClinicalBERT | 0.2190 |

Table 2: Architecture results as the encoder used varies

## 6 Discussion

Since there is no official baseline we can consider it as if it were the random classification, the F-1 macro of a random classifier proportional to the frequencies of the 9 classes is $F1_{random} = \frac{1}{9} = 0.\overline{11}$. Both methods are therefore acceptable for an initial approach to the task. Both architectures were completed after the competition had already concluded, but based on the final ranking, they would have finished $34^{th}$ and $33^{rd}$ respectively, considering a total of 172 participants and 563 total submissions.

The most obvious problem is visible in table 3, some classes are intrinsically difficult to classify, this could be caused both by the inhomogeneous distributions found in the traininig and test datasets, and by the great semantic closeness between them. The fact that there is a class (label 8) that includes all ambiguous cases suggests that the task is complicated even for an expert of the field, such complication and uncertainty is consequently greatly amplified in this context.

## 7 Conclusion

In this work, the task PsyDefDetect@BioNLP2026 was addressed by choosing to work with Encoder-only architectures, to verify whether it was possible to capture psycholinguistic patterns without resorting to Large Language Models. Two structurally opposed solutions have been proposed: a monolithic multi-task with distance loss and a hierarchical MIL-based, both built on MentalBERT. The two approaches achieve a test macro-F1 of 0.254 and 0.259, respectively, although far from the top positions in the rankings, the models learn a real signal. The most significant observation is that two such different architectures converge on nearly identical performance, the limitation being attributed more to the intrinsic separability of the data than to architectural choices. Consistently, the error is not widespread but localized in the same classes, indicating that the coarse structure of maturity of the defenses is grasped, while the finer nuances escape. A possible future approach could focus on the fusion of the two architectures, maintaining the hierarchical structure by making each node completely independent and finishing them by reusing attention mechanisms.

## 8 Links to external resources

- **Code repository**:

  `https://github.com/francesco-candoli/psydefdetect-bionlp2026`

- **Dataset**:

  hosted on the official CodaBench. Registration required.

  `https://www.codabench.org/competitions/12124/`

- **Shared task (PsyDefDetect @ BioNLP 2026)**:

  `https://psydefdetect-shared-task.github.io/`

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